$$V(x,y,z) = \frac{1}{4\pi\varepsilon_0} \int_V \frac{\rho(x,y,z)}{r'} \, dx\,dy\,dz$$

$$\vec{E}(x,y,z) = -\text{grad}\, V(x,y,z)$$

$$\text{rot}\,\vec{E} = -\text{rot grad}\, V$$

$$\Rightarrow \text{rot}\,\vec{E} = 0 \quad \text{(Elektrostatikte)}$$

$$\text{div grad}\, V = \Delta V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

silindirik: $\Delta V = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$

küresel: $\Delta V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial^2 V}{\partial \phi^2}$



$\Delta V = 0$ ← laplacian

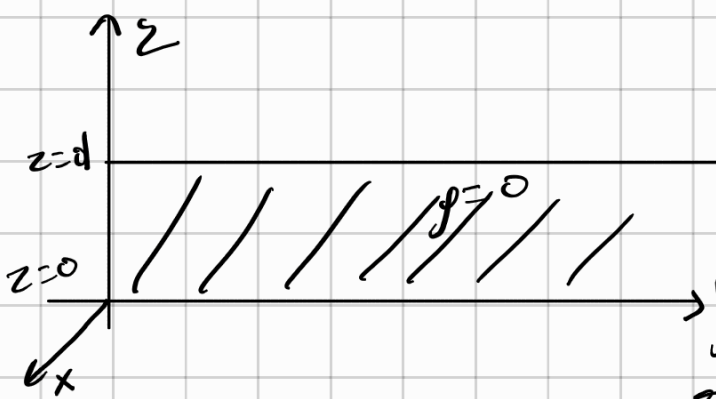
laplacian denklemi

Sayfa 26 · Pr 4

Pr.4. z = 0 ve z = d düzlemleri, sırasıyla, V = 0 ve V = V_0 potansiyellerine sahiptirler ve bu düzlemler arasındaki bölgede yük yoktur. Bu bölgede

$$V = \frac{V_0}{d} z, \qquad \mathbf{E} = -\frac{V_0}{d}\mathbf{e}_z$$

olduğunu gösteriniz.



Ara bölgede V ve $\vec{E}$ ifadeleri nedir

$\Delta V = 0$

değişmiyor

$\Delta V = 0$

$$\Delta V = \frac{\partial^2 V}{\partial x^2}^{0} + \frac{\partial^2 V}{\partial y^2}^{0} + \frac{\partial^2 V}{\partial z^2} = 0$$

$$\Rightarrow \frac{\partial^2 V}{\partial z^2} = 0 \xrightarrow{\text{integral}} \frac{\partial V}{\partial z} = A \Rightarrow V(z) = Az + B$$

$z = 0 \Rightarrow V(0) = 0$

$z = d \Rightarrow V(d) = V_0$

$V(0) = B = 0$

$V(d) = A\,d = V_0$

$A = \frac{V_0}{d}$

$\Rightarrow V(z) = \frac{V_0}{d} z$

$\vec{E} = -\text{grad}\,V = -\frac{\partial V}{\partial x}\vec{e}_x - \frac{\partial V}{\partial y}\vec{e}_y - \frac{\partial V}{\partial z}\vec{e}_z$ (0, 0)

$= -\frac{\partial}{\partial z}\left(\frac{V_0}{d} z\right)\vec{e}_z = -\frac{V_0}{d}\vec{e}_z$ //

S26. Pr5

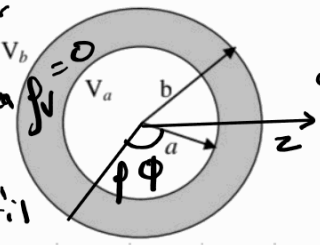
Pr.-5. Aynı eksenli, *a* ve b yarıçaplı, sonsuz uzun iki silindir arasındaki bölgede kaynak yoktur ve bu silindirler üzerinde potansiyel sabit V_a ve V_b değerlerine sahiptir. Potansiyel fonksiyonun açık ifadesini yazınız. Alan şiddetinin bir iletkenden diğerine nasıl değiştiğini gösteren eğriyi çiziniz.

Silindirik koordinatlar

ρ, ϕ, z → z'de sonsuza gider

ϕ ya bağımlı değil simetriktir

z'ye bağımlı değil



$a \le \rho \le b$

$V(\rho)$

$\rho = 0$ iken $\Delta V = 0$

$\Delta V = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0$ (0, 0)

gradyan

$= \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V(\rho)}{\partial \rho}\right) = 0$

$\Rightarrow \frac{\partial}{\partial \rho}\left(\rho \frac{dV(\rho)}{\partial \rho}\right) = 0$

$\Rightarrow \rho \frac{\partial V(\rho)}{\partial \rho} = A \Rightarrow \frac{\partial V(\rho)}{\partial \rho} = \frac{A}{\rho}$

$V(\rho) = A \ln \rho + B$ //

$\rho = a$ iken $V(a) = V_a$

$\rho = b$ iken $V(b) = V_b$

$\rho = a \Rightarrow A \ln a + B = V_a$

$\rho = b \Rightarrow -A \ln b + B = V_b$

$$A(\ln a - \ln b) = V_a - V_b$$

$$\Rightarrow \boxed{A = \frac{V_a - V_b}{\ln\left(\frac{a}{b}\right)}}$$

B yi bulalım

$$A \ln a + B = V_a$$

$$\frac{V_a - V_b}{\ln\left(\frac{a}{b}\right)} \cdot \ln a + B = V_a \Rightarrow B = V_a - \ln a \frac{V_a - V_b}{\ln\left(\frac{a}{b}\right)}$$

$$B = -\frac{V_a \ln b - V_b \ln a}{\ln\left(\frac{a}{b}\right)}$$

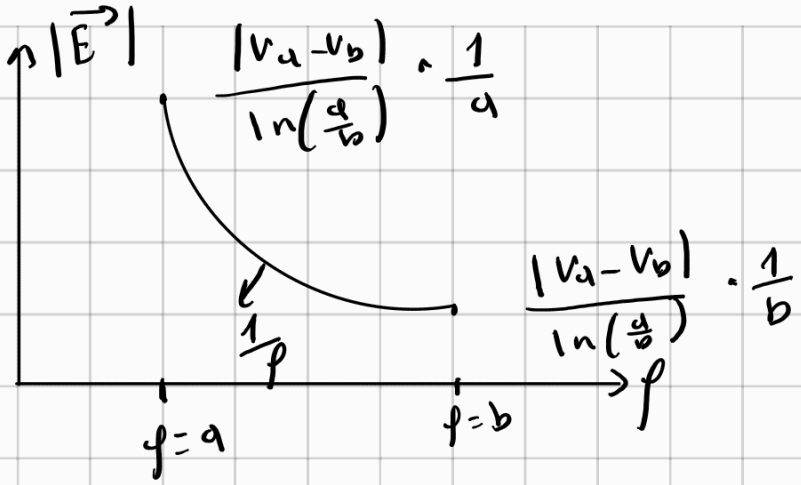
$$\boxed{V(\rho) = \frac{V_a - V_b}{\ln\left(\frac{a}{b}\right)} \ln \rho - \frac{V_a \ln b - V_b \ln a}{\ln\left(\frac{a}{b}\right)}}$$

$$\vec{E} = -\text{grad}\, V = -\frac{\partial V(\rho)}{\partial \rho} \vec{e}_\rho$$

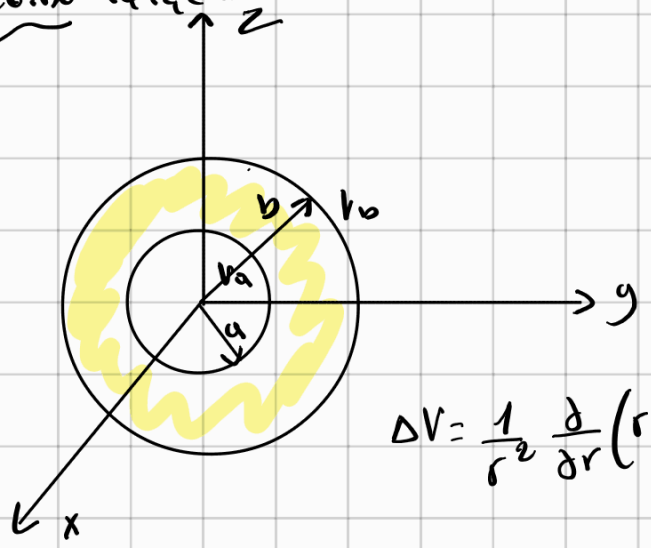
$$\vec{E}(\rho) = -\frac{1}{\rho} \frac{V_a - V_b}{\ln\left(\frac{a}{b}\right)} \vec{e}_\rho$$

kartezyen $\text{grad}\, V = \frac{\partial V}{\partial x} \vec{e}_x + \frac{\partial V}{\partial y} \vec{e}_y + \frac{\partial V}{\partial z} \vec{e}_z$

Silindirik $\text{grad}\, V = \frac{\partial V}{\partial \rho} \vec{e}_\rho + \text{-----}$



§26.Pb içiçe küre



Arada bölgede $a \leq r \leq b$ $V, \vec{E}$ için aldığımız

$\Delta V = 0 \Rightarrow$

$$\Delta V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right)$$

$$+ \frac{1}{r^2 \sin^2\theta}\frac{\partial V^2}{\partial \phi^2} = 0$$

r, ϕ, θ sadece r'ye bağlı,

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) = 0$$

$$\Rightarrow r^2 \frac{\partial V}{\partial r} = -A \text{ olsun}$$

$$\Rightarrow \frac{\partial V}{\partial r} = \frac{-A}{r^2} \Rightarrow V(r) = \frac{A}{r} + B$$

$r = a,\ V(a) = V_a \Rightarrow V_a = \frac{A}{a} + B$

$r = b,\ V(b) = V_b \Rightarrow V_b = \frac{A}{b} + B$

$$V_a - V_b = A\left(\frac{1}{a} - \frac{1}{b}\right)$$

$$A = \frac{V_a - V_b}{\frac{1}{a} - \frac{1}{b}}$$

$$V_a = \frac{A}{a} + B \Rightarrow V_a = \frac{1}{a}\,\frac{V_a - V_b}{\left(\frac{1}{a} - \frac{1}{b}\right)} + B$$

$$B = V_a - \frac{1}{a}\,\frac{V_a - V_b}{\frac{1}{a} - \frac{1}{b}}$$

$$= V_a - \frac{ab}{ab}\,\frac{V_a - V_b}{b - a}$$

$$B = \frac{V_a(b-a) + b(V_a - V_b)}{b-a}$$

$$B = \frac{bV_b - aV_a}{b-a} = \frac{aV_a - bV_b}{a-b}$$

$$\boxed{V(r) = \frac{V_a - V_b}{\frac{1}{a} - \frac{1}{b}}\,\frac{1}{r} + \frac{aV_a - bV_b}{a-b}}$$

$$\vec{E}(r) = -\text{grad}\,V = -\frac{\partial V(r)}{\partial r}\,\vec{e}_r$$

$$= -\frac{\partial}{\partial r}\left(\frac{V_a - V_b}{\frac{1}{a} - \frac{1}{b}}\,\frac{1}{r}\right)\vec{e}_r$$

$$= \frac{V_a - V_b}{\frac{1}{a} - \frac{1}{b}}\,\frac{1}{r^2}\,\vec{e}_r$$

S.26 Pr.9

olduğunu gösteriniz (Gauss özdeşliği).

Pr.-9. Bir elektrostatik alanda

$$\mathbf{E} = A\frac{\cos\theta}{r^3}\mathbf{e}_r + B\frac{\sin\theta}{r^3}\mathbf{e}_\theta$$

olduğu gözlenmiştir (A,B = sabit).

a) A ile B arasındaki ilişkiyi bulunuz.
b) Potansiyel fonksiyonunu belirleyiniz.

a) $\text{rot}\,\vec{E} = 0$ ♡

$$\vec{E} = E_r\vec{e}_r + E_\theta\vec{e}_\theta + E_\phi\vec{e}_\phi$$

$$\Rightarrow E_\phi = 0,\quad E_r = A\frac{\cos\theta}{r^3},\quad E_\theta = B\frac{\sin\theta}{r^3}$$

$$\text{rot}\,\vec{E} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta\,E_\phi) - \frac{\partial E_\theta}{\partial\phi}\right]\vec{e}_r$$

$$+ \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial E_r}{\partial\phi} - \frac{\partial}{\partial r}(rE_\phi)\right]\vec{e}_\theta$$

$$+\frac{1}{r}\left[\frac{\partial}{\partial r}(rE_\theta)-\frac{\partial E_r}{\partial \theta}\right]\vec{e}_\theta=0$$

$$\frac{1}{r}\left[\frac{\partial}{\partial r}(rE_\theta)-\frac{\partial E_r}{\partial \theta}\right]\vec{e}_\varphi=0$$

$$\Rightarrow \frac{\partial}{\partial r}(rE_\theta)=\frac{\partial E_r}{\partial \theta}$$

$$\frac{\partial}{\partial r}\left(rB\frac{\sin\theta}{r^3}\right)=\frac{\partial}{\partial \theta}\left(A\frac{\cos\theta}{r^3}\right)$$

$$-2B\frac{\sin\theta}{r^3}=-\frac{A\sin\theta}{r^3}$$

$$\boxed{A=2B}$$

b) $\vec{E}=-\text{grad}V$

$$=-\frac{\partial V}{\partial r}\vec{e}_r-\frac{1}{r}\frac{\partial V}{\partial \theta}\vec{e}_\theta-\underbrace{\frac{1}{r\sin\theta}\frac{\partial V}{\partial \varphi}e_\varphi}_{0}$$

$$E_r\vec{e}_r+E_\theta\vec{e}_\theta=-\frac{\partial V}{\partial r}\vec{e}_r-\frac{1}{r}\frac{\partial V}{\partial \theta}\vec{e}_\theta$$

$\boxed{E_r=-\frac{\partial V}{\partial r}}$, $E_\theta=-\frac{1}{r}\frac{\partial V}{\partial \theta}$ $\quad V(r,\theta)$

bunu seçeceğim

$$V=-\int E_r\,dr+\underbrace{f(\theta)}_{\text{thetaya bağımlılığın}}$$

$$=-\int\frac{A}{r^3}\cos\theta\,dr+f(\theta)$$

$$=\frac{A}{2r^2}\cos\theta+f(\theta)=V$$

$$E_\theta=-\frac{1}{r}\frac{\partial V}{\partial \theta}\Rightarrow B\frac{\sin\theta}{r^3}=-\frac{1}{r}\frac{\partial}{\partial \theta}\left(\frac{A}{2r^2}\cos\theta+f(\theta)\right)$$

$$\rightarrow \frac{B\sin\theta}{r^2} = +\frac{A}{2r^2}\sin\theta - \frac{\partial f(\theta)}{\partial\theta}$$

Önceden $A=2B$ olduğunu biliyorduk

$$\frac{A\sin\theta}{2r^2} = \frac{A}{2r^2}\sin\theta - \frac{\partial f(\theta)}{\partial\theta}$$

$$\rightarrow \frac{\partial f(\theta)}{\partial\theta} = 0 \Rightarrow f(\theta) = C$$

$$V = \frac{A}{2r^2}\cos\theta + f(\theta) = \frac{A}{2r^2}\cos\theta + C$$

$$\vec{r}\rightarrow\infty \Rightarrow \vec{E}\rightarrow 0 \Rightarrow \vec{V}\rightarrow 0$$

$$\lim_{r\rightarrow\infty} V = 0 + C = 0 \Rightarrow C=0$$